

ME ANNALS

DEPARTMENT OF MECHANICAL ENGINEERING



2023-24

MISSION & VISION



VISION

To achieve global standards in value based education in Mechanical Engineering.

MISSION

M1: To impart high quality application based education to make the students globally competitive.

M2: To foster human values, national ethos, and compassion for environment and society.

M3: To create a productive atmosphere in the department with the motto - 'Sab ka sath, sabka vikas'.

MECHANICAL ENGINEERING



The Mechanical Engineering is equipped well with the well qualified and vibrant professional faculty's dedicated for the betterment of students.

The mechanical Engineering department at SRMCEM has analysis, design, manufacturing and maintenance of mechanical systems as its main functional areas.

The department offers a practical learning experience in the field of heat, fluid, energy, and motion in a mechanical system.

The students of the department have a reputation of makers and problem solvers.

The Mechanical Engineering department of SRMCEM has an excellent faculty base with core competence in Mechanics, Kinematics, Material Sciences and Structural analysis.

The department uses INVENTOR and SOLID WORK software which help students in learning designing, manufacturing, and drafting processes.

An undergraduate degree in Mechanical Engineering from SRMCEM means a solid foundation in the art and science of mechanical engineering through which the ME students can easily compete with the best in their profession.

The various teaching methodologies along with practical knowledge is imparted to the students in the department through industrial visits, Guest Lectures, Invited Talks, Workshops and Seminars.

EDITOR'S NOTE

Welcome to the **ME ANNALS** from the **Mechanical Engineering** department. We're excited to bring you the latest news and developments from our department.

In this, you'll find a range of articles and features that showcase the achievements and contributions of our faculty, staff, and students.

The constant contribution of students and faculty members made the completion of this magazine a success and I would like to thank them. A special thanks to all the supporting teams and their members.

We hope you enjoy reading this issue and learning more about our department's innovative and impactful work. As always, we welcome your feedback and suggestions for future topics.

Best regards,

Editor-Chief

Er Sudhir Kr. Katiyar

Asst.Professor

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Designing of electrochemical cut-off grinding process of Ti–6Al–4V

The electrochemical cut-off grinding (ECCG) process is a chemical energy-based hybrid machining process in which material is removed from the workpiece by anodic dissolution along with mechanical abrasive action. Ti–6Al–4V is the most demanding material to make the components in aircraft and aerospace industries due to their exceptional properties such as high strength-to-weight ratio, high-temperature resistance, high fatigue resistance, and unique corrosion resistance. In the present study, an electrochemical cut-off grinding setup has been designed and developed in-house with the consideration of the size and weight of different parts along with special consideration of corrosion resistance. The experimental setup has been used to investigate the influence of process parameters namely voltage (V), tool feed rate (TFR), electrolyte concentration (EC), electrolyte flow rate (EFR), and grinding wheel rotation (RPM) on response parameters i.e., material removal rate (MRR) and surface roughness (Ra) during ECCG process of Ti–6Al–4V. The experiments are conducted by using a metal-bonded diamond grinding wheel (D151) and copper wheel respectively both of the same sizes, and the results are compared to know the effect of mechanical grinding action along with ion dissolution. The experimental result shows that tool feed rate, electrolyte flow rate, and grinding wheel rotation are the most efficient process parameters which improve the material removal rate and surface finish both simultaneously.

The most suitable range of tool feed rate, electrolyte flow rate, and grinding wheel rotation are 0.09–0.15 mm/min, 5–7 l/min, and 800–1600 rpm respectively. The outcome of the comparative analysis of a metal-bonded diamond grinding wheel and copper grinding wheel points out that material removal rate and surface finish improve up to 8.4% and 8.6% respectively by using a diamond grinding wheel.



Er. Sunil Kumar Yadav

Assist. Professor

Department Mechanical Engineering

Experimental investigation of electrochemical cut-off grinding for Inconel-718 and comparative study with Ti-6Al-4V alloy

In the present study, a hybrid machining process known as an electrochemical cut-off grinding (ECCG) is used for grinding of Inconel-718 material. The specimen (superalloy) contains nickel as major component. It cannot be machined conventionally due to its outstanding properties. During the electrochemical grinding of Inconel 718, the author examines the effects of input voltage (v), tool feed rate (mm/min), electrolytic concentrations (g/ltr), electrolytic flow rate (ltr/min), and grinding speed (rpm) on responses such as material removal rate (MRR) and surface roughness (Ra). The tests contains the bronze-bonded diamond abrasive grinding wheel. The MRR and surface roughness of Inconel 718 and titanium super alloy during the ECCG process are also compared in this research project. Based on the findings of the studies, it appears that the tool feed rate, the flow rate of the electrolyte, and the rotation of the grinding wheel are the maximum influential parameters for enhancing MRR and surface finish during electrochemical cut-off grinding of Inconel 718. Tool feed rates of 0.09 mm/min to 0.18 mm/min, electrolyte flow rates of 5 ltr/min to 8 ltr/min, and grinding wheel speeds of 800 rpm to 2000 rpm are the ideal influencing factors.

The comparative analysis demonstrates that Ti-6Al-4 V alloy cannot match Inconel 718's machinability. Compared to Ti-6Al-4 V alloy, Inconel-718 has an average material removal rate that is up to 1.73times higher and an average surface quality that is up to 3.16 times better.



Er. Sunil Kumar Yadav

Assist. Professor

Department Mechanical Engineering

STUDENT ACHIEVEMENTS



STUDENT ACHIEVEMENTS

STUDENT NAME	EVENT
ABHINAV JAISWAL	PARTICIPATED IN ROBORACE & ROBOSOCCKER GANTAVYA 2K22 ORGANISED AT SRMCEM
AYUSH NIGAM	PARTICIPATED IN LINE FOLLOWER GANTAVYA 2K22 ORGANISED AT SRMCEM
ABHISHEK KUMAR UPADHAYAY	IIT DHANBAAD ROBOWAR & ROBORACE <u>1ST, 2ND & 3RD POSITION</u> IIT JODHPUR ROBOWAR & ROBORACE <u>1ST & 2ND POSITION</u> AKTU ZONALS ROBOWAR & ROBORACE <u>1ST POSITION</u> AKTU STATES ROBOWAR & ROBORACE <u>1ST POSITION</u> PARTICIPATED IN ROBORACE, ROBOSOCCKER & AQUARUSH IN <u>CELESTA</u> (IIT PATNA)
AKHIL TRIPATHI	BBDU LUCKNOW ROBOWAR <u>FIRST PRIZE</u> MUIT LUCKNOW ROBOWAR <u>FIRST PRIZE</u> PARUL UNIVERSITY VADODARA ROBORACE & ROBOWAR <u>2ND PRIZE</u> SRMU LUCKNOW ROBOWAR & ROBORACE <u>1ST & 2ND PRIZE</u> BPIT DELHI ROBOWAR <u>RUNNER-UP</u>
PANKAJ SINGH	SECURED 2ND POSITION DEPARTMENTAL QUIZ IN GANTAVYA 2K22 ORGANISED AT SRMCEM
PRATHAK MISHRA	SECURED 1ST POSITION DEPARTMENTAL QUIZ IN GANTAVYA 2K22 ORGANISED AT SRMCEM

STUDENT ACHIEVEMENTS

STUDENT NAME	EVENT
ALOK KUMAR	BBDU LUCKNOW ROBOWAR FIRST PRIZE MUIT LUCKNOW ROBOWAR FIRST PRIZE PARUL UNIVERSITY VADODARA ROBORACE & ROBOWAR 2ND PRIZE SRMU LUCKNOW ROBOWAR & ROBORACE 1ST & 2ND PRIZE BPIT DELHI ROBOWAR RUNNER-UP
GOPAL KRISHNA SRIVASTAV	IIT DHANBAAD ROBOWAR & ROBORACE 1ST,2ND & 3RD POSITION IIT JODHPUR ROBOWAR & ROBORACE 1ST& 2ND POSITION AKTU ZONALS ROBOWAR & ROBORACE 1ST POSITION AKTU STATES ROBOWAR & ROBORACE 1ST POSITION BPIT DELHI ROBO SOCCER 1ST& 2ND POSITION
ANKIT KUMAR MISHRA	BBDU LUCKNOW ROBOWAR FIRST PRIZE MUIT LUCKNOW ROBOWAR FIRST PRIZE IIT PATNA ROBORACE & ROBOWAR 2ND PRIZE SRMU LUCKNOW ROBOWAR & ROBORACE 1ST & 2ND PRIZE PSIT KANPUR ROBORACE & ROBOSOCCER 1ST & 2ND PRIZE BPIT DELHI ROBOWAR RUNNER-UP

PLACEMENTS

Student's Name: Aditi <u>Sonker</u> Company Name: 1-IndoPlast. 2-Learning Shala. 3-Gravity.	
Student's Name: <u>Amitanshu</u> Gupta Company Name: KP Reliable	
Student's Name: Abhishek Kumar <u>Upadhayay</u> Company Name: Gravity classes	
Student's Name: Amit Pandey Company Name: KP Reliable	
Student's Name: <u>Anshuman</u> Tripathi Company Name: RDC Concrete	

PLACEMENTS

Student's Name: Chandra Pratap Mall Company Name: KP Reliable	
Student's Name: Gopal Krishna Srivastava Company Name: KP Reliable	
Student's Name: Parvez Alam Company's Name: 1. INDOPLAST 2. KP RELIABLE	
Student's Name: Priyanshu Mishra Company Name: K.P Reliable	
Student's Name: Sameer Singh Company Name: K.P Reliable	 Sameer Singh

SPORTS

Sports occupies a vital role at Shri Ramswaroop Memorial Group of Professional Colleges. The sports activities at the SRMGPC campus provide the students with a proper physique and a positive mentality

STUDENT NAME	SPORT	EVENT
Anchal Srivastava	Kabaddi	Spardha 2023 Runner Up
	Cricket	Spardha 2023 Participated Spardha 2022 Winner SRMPL 2022 Participated SRMPL 2023 Participated
Manas Dubey	Kabaddi	Spardha 2023 Runner Up
	Cricket	Spardha 2023 Participated Spardha 2022 Winner SRMPL 2022 Participated SRMPL 2023 Participated Jaipuriya Tournament Silver
Archit Dubey	Basket Ball	AKTU Zonal 2021 Gold AKTU Zonal 2022 Gold State Semi Finalist 2021 State Semi Finalist 2022 AKTU Zonal Semi Finalist 2023
Pankaj Singh	Kabaddi	Spardha 2023 Runner Up

STUDENT NAME	SPORT	EVENT
Rajvardhan Singh	Cricket	Spardha 2023 Participated SRMPL 2023 Participated
Sujeet Singh	Cricket	Spardha 2023 Participated SRMPL 2023 Participated
Anshuman Tripathi	Kho-Kho	AKTU ZONAL 2022 Gold State Level 2022 participated Spardha 2023 Runner Up
Vansh Kumar Kushwaha	Kho-Kho	AKTU ZONAL 2022 Gold
Prakhar Yadav	Kho-Kho	AKTU ZONAL 2022 Gold State Level 2022 participated Spardha 2023 Runner Up AKTU Zonal 2023 Runner Up
Pankaj Singh	Kabaddi	Spardha 2023 Runner Up